EAST WEST UNIVERSITY

Project Report

"ReBite-A Smart Marketplace for Surplus Food Recovery"

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1.Abstract

ReBite is a proposed AI-powered digital marketplace designed to connect restaurants, cafes, bakeries, supermarkets, and event organizers with price-conscious consumers seeking affordable, safe, surplus food. Large volumes of edible food are discarded daily due to overproduction and inaccurate demand forecasting, while students, low-income families, and budget-conscious consumers struggle with rising food costs. ReBite addresses both sides of this gap through a four-sided platform (Customer Hub, Restaurant Hub, NGO/Donation Workflow, and Admin & Billing Hub) built around an AI waste-prediction engine, a countdown-based dynamic pricing ladder, an NGO donation channel for unsold food, a Student Saver Membership, and carbon-impact tracking.

This report documents the complete project planning package developed for ReBite under the CSE 495 IT Project Management & Entrepreneurship course. It consolidates the project overview, requirements, stakeholder analysis, value proposition, SWOT analysis, a four-level Work Breakdown Structure spanning an 889-day (≈16-month) schedule, a ten-item sprint-based risk register, a labor cost and contingency-reserve analysis totalling approximately BDT 3,277,460, and the design iterations made in response to instructor feedback. The report is intended as a single reference document for project governance, planning review, and academic evaluation.

2.Introduction

Food waste is one of the most overlooked economic and environmental challenges facing urban food systems today. In Bangladesh, as in many developing economies, restaurants, bakeries, and supermarkets routinely discard significant volumes of unsold but perfectly edible food because of overproduction, inaccurate demand forecasting, and short shelf life, while a large segment of the population — particularly students and low-income households — continues to struggle with rising food costs and limited access to affordable, quality meals.

ReBite is conceived as an AI-powered surplus food marketplace that bridges this gap. Unlike conventional food-delivery platforms such as Foodpanda or Foodi, which focus purely on convenience and full-price delivery, ReBite is purpose-built around food-waste reduction, affordability, sustainability, and measurable social impact. Verified restaurants, cafés, bakeries, and supermarkets list surplus food at automatically discounted prices as the pickup deadline approaches; consumers reserve and pay for the food online and collect it in person through QR-code verification; and any food that ultimately remains unsold can be routed to partner NGOs instead of being thrown away.

This report presents the full academic project-management package built around the ReBite concept — from the original project proposal through to a revised Work Breakdown Structure, a

sprint-based risk register, a detailed labor-cost and contingency analysis, and the concrete changes made to the plan in response to instructor and stakeholder feedback.

3.Problem Statement

Food waste has become a major economic and environmental issue, and the current market fails to address it in a structured, incentive-aligned way.

Current Challenges

- Restaurants and food businesses lose revenue from unsold, discarded food.
- Consumers — especially students and low-income families — face increasing food costs.
- Large amounts of safe, edible food end up in landfills, contributing to environmental harm.
- Existing food-delivery platforms (e.g., Foodpanda, Foodi) are optimized for convenience and full-price delivery, not waste reduction.
- There is no dedicated, large-scale surplus-food marketplace operating in Bangladesh today.

Consequences

- Direct financial loss for restaurants, cafés, bakeries, and supermarkets.
- Increased carbon emissions from food production, transport, and landfill decomposition.
- Wasted resources — water, labor, packaging, and energy — invested in food that is never consumed.
- Reduced food accessibility and affordability for students and low-income groups.

4.Project Overview & Proposed Solution

ReBite provides a digital marketplace where verified food businesses can list surplus food at discounted prices, and where consumers can discover, reserve, and pick up that food conveniently and safely. The platform is organized around four interconnected hubs — Customer, Restaurant, NGO/Donation, and Admin & Billing — supported by an AI waste-prediction engine and a dynamic, countdown-based pricing model.

What Customers Can Do

- Discover nearby surplus food listings in real time.
- Reserve food instantly and pay online.
- Collect their order through QR-code pickup verification.
- Save money on quality meals while contributing to sustainability.

What Businesses Can Do

- Recover revenue that would otherwise be lost to unsold inventory.
- Reduce food waste and associated disposal costs.
- Gain access to new, price-sensitive customer segments.
- Use AI-driven analytics to forecast demand and optimize inventory.

Project Scope (as planned)

The project scope, as reflected in the revised Work Breakdown Structure, spans project kickoff and planning, UX and system design, parallel development of the four platform hubs, integration and testing, and deployment and closure — an end-to-end delivery plan covering roughly 16 months of calendar time from kickoff to project closure.

5.Project Objectives

- Design and specify a four-sided digital marketplace (Customer, Restaurant, NGO, Admin) that reduces commercial food waste in Bangladesh.
- Deliver an AI-assisted waste-prediction and dynamic-pricing system that increases the likelihood surplus food is sold before spoilage.
- Provide a verified, trust-based transaction environment (freshness timestamps, hygiene badges) so consumers can safely buy discounted near-expiry food.
- Create a sustainable, diversified revenue model that supports both business viability and continued food-waste reduction incentives.
- Establish a formal donation pathway so surplus food that is not sold can still reach NGOs rather than landfill.
- Produce a realistic, governance-ready project plan — WBS, schedule, risk register, and cost/contingency budget — suitable for execution and academic evaluation.

6.Value Proposition

For Consumers

- Affordability: Access to quality meals at a significant discount as pickup deadlines approach.
- Trust & Safety: Food is sourced only from verified restaurants with visible freshness/hygiene badges.
- Impact & Identity: Every purchase shows meals rescued and estimated CO2 savings, reframing the purchase as a 'meal rescued' rather than 'leftover food'.

For Restaurants & Food Businesses

- Loss Recovery: Selling surplus at a discount recovers revenue that would otherwise be a total loss (0 BDT) from thrown-out food.
- Smarter Inventory: AI-driven waste prediction and analytics help businesses plan inventory more accurately over time.
- Low-Friction Revenue: A low, predictable flat platform fee plus optional premium subscriptions keeps the cost of participation transparent.

For NGOs

- Reliable Supply: A structured, scheduled donation channel to receive surplus food that is not sold, with pickup-window tracking.

For Society & the Environment

- Sustainability: Reduced landfill waste, lower carbon emissions, and improved food access for students and low-income groups create measurable social and environmental value.

7.Target Market

ReBite serves a three-tier market structure spanning direct consumers, supplying businesses, and mission-aligned institutional partners.

Segment	Description
Primary Market	University students, young professionals, and budget-conscious consumers seeking affordable meals.
Secondary Market	Restaurants, cafés, bakeries, supermarkets, and event organizers with recurring surplus food.
Tertiary Market	NGOs, sustainability organizations, and government initiatives focused on food security and waste reduction.

8.Stakeholder Analysis

The ReBite ecosystem involves eight primary stakeholder groups spanning consumer, business, institutional, and academic interests.

Stakeholder	Role / Interest	Key Expectations
Consumers (students, professionals, budget-conscious households)	End users who browse, reserve, and purchase surplus food.	Affordable pricing, food safety and freshness, easy pickup, transparent impact tracking.
Restaurants, cafés, bakeries, supermarkets	Supply-side merchants who list surplus food and manage listings.	Fair fees, additional revenue recovery, simple onboarding, useful analytics.
Partner NGOs	Recipients of donated surplus food that is not sold in time.	Reliable donation scheduling, predictable pickup windows, transparent impact reporting.
Platform Admin / Operations Team	Manages users, restaurants, orders, listings, donations, billing, and system health.	Reliable tools, clear audit trails, low-friction dispute resolution.
Project Team (developers, PM, UX, QA)	Designs, builds, tests, and delivers the platform.	Clear requirements, realistic timeline, manageable risk exposure.
Course Instructor / University (East West University)	Academic evaluator and project sponsor for CSE 495.	A credible, well-governed, professionally documented project plan.
Government & Regulatory Bodies	Oversight of food-safety and consumer-protectio n compliance.	Adherence to hygiene, safety, and data-privacy regulations.

Future Investors / Sustainability Partners	Potential funders or CSR partners interested in scaling the platform.	Clear business model, growth potential, measurable ESG/impact metrics.

9.Requirements Specification

9.1 Functional Requirements

Customer Hub

- User registration, login, and profile management.
- Search and discovery of nearby surplus food listings.
- Food reservation and Surprise Food Box selection.
- Online payment processing.
- QR-code based pickup verification.
- Ratings and reviews for completed orders.
- Carbon/impact dashboard showing meals rescued, waste reduced, and CO2 saved.
- Student Saver Membership subscription and benefits.

Restaurant Hub

- Restaurant registration and hygiene/trust verification.
- Surplus food listing upload with freshness timestamp and badge.
- AI waste-prediction engine for demand and surplus forecasting.
- Dynamic, countdown-based pricing engine with a merchant-defined price floor.
- Restaurant analytics dashboard and sales/earnings reports.

NGO / Donation Workflow

- NGO partner onboarding and verification (admin-managed).
- Donation request creation and management.
- Pickup scheduling with defined time windows.
- Food distribution tracking through to collection.

Admin & Billing Hub

- User and restaurant account management.
- Platform-wide order management console.
- Flat-fee billing and subscription engine.

- Featured-listing placement engine.
- Sustainability reports (paid B2B product).
- System monitoring and platform-wide analytics dashboards.

9.2 Non-Functional Requirements

- Security: Security: end-to-end encryption and role-based access control for all personal, location, and payment data.
- Reliability: Reliability: the payment flow must support at least two gateway providers plus a cash-on-pickup fallback to minimize checkout failures.
- Performance: Performance: search, listing, and pricing updates should reflect near real-time inventory changes as countdown discounts trigger.
- Scalability: Scalability: the architecture must support parallel onboarding of restaurants, NGOs, and consumers across multiple cities.
- Usability: Usability: mobile-first design for both the consumer and restaurant apps, validated through wireframes and usability review.
- Auditability: Data Integrity: billing and revenue calculations must be reconciled against manual calculations before go-live, with an audit log for every billing event.
- Compliance: Compliance: hygiene and food-safety verification badges must be enforced before a restaurant listing goes live.

10.System Modules

The platform is composed of four functional hubs, each aligned to a distinct stakeholder group and its corresponding workflow.

Hub	**Key Features**
Customer Hub	Registration & login; search nearby surplus food; food reservation; online payment; QR pickup; ratings & reviews; carbon impact dashboard; Student Saver Membership.
Restaurant Hub	Restaurant verification; upload surplus food; AI waste prediction; dynamic pricing control; analytics dashboard; sales reports.
NGO Hub (Admin-managed	Donation requests; pickup scheduling; food distribution tracking.

Donation Workflow)	
Admin Hub	User management; restaurant verification; complaint management; analytics; system monitoring; billing & revenue reports.

11.Unique Selling Propositions

1. AI Waste Prediction System

ReBite analyzes historical sales data to predict daily demand, expected surplus, and waste trends — improving inventory management and reducing overproduction.

2. Smart Dynamic Pricing Engine

Discounts increase automatically as the pickup deadline approaches: 20% off with 3 hours remaining, 50% off with 1 hour remaining, and 80% off with 30 minutes remaining — maximizing the chance surplus food is sold rather than wasted.

3. NGO Food Donation Network

Unsold food can be routed directly to partner NGOs, creating measurable social impact and CSR opportunities for participating restaurants.

4. Student Saver Membership

A subscription model offering additional discounts, priority reservations, and campus-exclusive offers — designed to drive recurring revenue and strong student adoption.

5. Carbon Impact Tracking

Every purchase shows meals rescued, food waste reduced, and estimated CO2 emissions saved, reinforcing user engagement and environmental awareness.

6. Surprise Food Box

Restaurants can bundle surplus items into mystery food boxes for fast inventory clearance and an engaging customer experience.

12.Competitor Analysis

ReBite's positioning is deliberately differentiated from general-purpose food-delivery apps, which compete on convenience rather than waste reduction or sustainability.

Feature	**Foodpanda**	**Foodi**	**ReBite**
Food Delivery	✓	✓	✗
Surplus Food Marketplace	✗	Partial	✓
AI Waste Prediction	✗	✗	✓
NGO Donation	✗	✗	✓
Student Membership	✗	✗	✓
Carbon Tracking	✗	✗	✓

13.SWOT Analysis

A SWOT analysis was developed for ReBite based on the proposal, competitor comparison, and the project's own risk register, to give a balanced strategic view of the venture.

STRENGTHS	**WEAKNESSES**
• First-of-its-kind, dedicated surplus-food marketplace in Bangladesh — no direct large-scale competitor. • AI-driven waste prediction and dynamic pricing engine differentiate ReBite from convenience-only delivery apps. • Diversified revenue streams: commission, subscriptions, featured listings, sustainability reports. • Strong social-impact and CSR appeal (NGO donation network, carbon tracking).	• New, unproven market — no existing large-scale local benchmark to validate demand. • AI predictions and dynamic pricing depend on sufficient historical sales data, which new restaurants will initially lack. • Revenue model (flat fee, price floors) is still an untested hypothesis pending real transaction data. • Platform trust must be earned quickly — a single food-safety

• Multi-sided value creation: benefits consumers, restaurants, NGOs, and the environment simultaneously.	incident could cause lasting brand damage. • Pickup-only model (no delivery) may limit convenience relative to established delivery apps.
OPPORTUNITIES • Rising sustainability awareness and ESG expectations among consumers, businesses, and investors. • Potential partnerships with NGOs, sustainability organizations, and government food-security initiatives. • Expansion beyond an initial city/campus base into other urban centers of Bangladesh. • Growing smartphone and digital-payment penetration supports platform adoption. • Adjacent growth into corporate cafeterias, catering surplus, and university dining halls.	**THREATS** • Established delivery platforms (Foodpanda, Foodi) could add competing surplus-food features. • Food-safety incidents traced to a listing could damage trust irreversibly (see Risk RSK-03). • Regulatory and food-hygiene compliance requirements may evolve or tighten. • Restaurant resistance to discounting or onboarding into a donation-first rollout (see Risk RSK-10). • Payment gateway reliability and transaction-failure risk at checkout (see Risk RSK-04).

14. Work Breakdown Structure & Project Timeline

The project runs from kickoff to closure over 8 Level-2 phases and 889 total person/calendar-days of planned work (roughly 16 months of elapsed calendar time), further decomposed to Level 4 activities for estimation, assignment, and tracking. Project start: 27 July 2026. Planned project end: 26 November 2027.

14.1 Phase Summary (Level 2)

WBS ID	Phase (Level 2)	Duration	Start	End
1.0	Project Kickoff & Planning	58d	27 Jul 2026	22 Sep 2026
2.0	UX & System Design	106d	23 Sep 2026	6 Jan 2027

3.0	Module A: Customer Hub	160d	7 Jan 2027	15 Jun 2027
4.0	Module B: Restaurant Hub	157d	7 Jan 2027	12 Jun 2027
5.0	Module C: NGO / Donation Workflow	91d	7 Jan 2027	7 Apr 2027
6.0	Module D: Admin & Billing Hub	153d	7 Jan 2027	8 Jun 2027
7.0	Integration & Testing	93d	16 Jun 2027	16 Sep 2027
8.0	Deployment & Closure	71d	17 Sep 2027	26 Nov 2027

Table 14.1 — WBS Level-2 phase roll-up (durations and dates roll up from Level 3/4 child activities).

14.2 Work Package Detail (Level 3)

Each Level-2 phase is decomposed into the following Level-3 work packages; each of these is further broken into Level-4 activities in the master WBS workbook for day-to-day estimation and tracking.

WBS ID	Work Package (Level 3)	Duration	Owner
1.1	Kickoff meeting	3d	Workstream Lead
1.2	Requirements gathering & stakeholder interviews	13d	Workstream Lead
1.3	Finalize revenue & pricing model (flat fee, price floors)	12d	Workstream Lead
1.4	Project plan sign-off	30d	Workstream Lead
2.1	Wireframes & UI/UX design (customer + restaurant apps)	28d	Workstream Lead
2.2	Database schema design	5d	Workstream Lead
2.3	System architecture & API design	18d	Workstream Lead
2.4	Trust & freshness badge system spec	12d	Workstream Lead

2.5	Dynamic pricing engine design (countdown tiers + merchant floor)	13d	Workstream Lead
2.6	Design review & sign-off	30d	Workstream Lead
3.1	Registration & login	14d	Workstream Lead
3.2	Search & discovery (nearby surplus food)	16d	Workstream Lead
3.3	Reservation & Surprise Food Box flow	15d	Workstream Lead
3.4	Online payment integration	22d	Workstream Lead
3.5	QR pickup verification	12d	Workstream Lead
3.6	Ratings & reviews	9d	Workstream Lead
3.7	Impact dashboard (meals rescued & money saved)	12d	Workstream Lead
3.8	Student Saver Membership	15d	Workstream Lead
3.9	Unit testing: Customer Hub	15d	Workstream Lead
3.10	Module A review & sign-off	30d	Workstream Lead
4.1	Restaurant registration & hygiene verification	17d	Workstream Lead
4.2	Surplus listing upload (freshness timestamp + badge)	15d	Workstream Lead
4.3	AI waste prediction engine (prototype-ready)	29d	Workstream Lead
4.4	Dynamic pricing engine (countdown ladder + price floor)	20d	Workstream Lead
4.5	Restaurant analytics dashboard (premium tier)	19d	Workstream Lead
4.6	Sales & earnings reports	12d	Workstream Lead
4.7	Unit testing: Restaurant Hub	15d	Workstream Lead

4.8	Module B review & sign-off	30d	Workstream Lead
5.1	NGO partner onboarding & verification (admin-managed)	12d	Workstream Lead
5.2	Donation request workflow	15d	Workstream Lead
5.3	Pickup scheduling	10d	Workstream Lead
5.4	Food distribution tracking	11d	Workstream Lead
5.5	Unit testing: NGO / Donation Workflow	13d	Workstream Lead
5.6	Module C review & sign-off	30d	Workstream Lead
6.1	User & restaurant management console	17d	Workstream Lead
6.2	Platform-wide Order Management console	14d	Workstream Lead
6.3	Flat-fee billing & subscription engine	21d	Workstream Lead
6.4	Featured listing placement engine	13d	Workstream Lead
6.5	Sustainability reports (B2B paid product)	14d	Workstream Lead
6.6	System monitoring dashboard	12d	Workstream Lead
6.7	Platform-wide analytics dashboard	17d	Workstream Lead
6.8	Unit testing: Admin Hub	15d	Workstream Lead
6.9	Module D review & sign-off	30d	Workstream Lead
7.1	Integrate all modules (A+B+C+D)	24d	Workstream Lead
7.2	System testing	25d	Workstream Lead
7.3	Security & payment testing	20d	Workstream Lead
7.4	User acceptance testing (UAT)	24d	Workstream Lead
8.1	Deploy to production	12d	Workstream Lead
8.2	Restaurant onboarding & training	15d	Workstream Lead

8.3	Documentation & handover	14d	Workstream Lead
8.4	Project closure	30d	Workstream Lead

Table 14.2 — WBS Level-3 work packages by phase.

14.3 Scheduling Notes & Assumptions

- Durations are calendar days (not business days); the team may wish to convert to a business-day calendar during execution.
- Formal sign-offs (project plan, design, and each module) are planned as 30-calendar-day review/approval windows to allow realistic stakeholder review, revisions, and re-approval.
- Customer, Restaurant, NGO/Donation, and Admin workstreams (Modules A–D) run in parallel after Design Sign-off and before Integration & Testing begins.
- The NGO Hub is treated as an admin-managed donation workflow rather than a standalone user-facing hub, reflecting the prototype's actual screen structure.

15. Risk Management

Risk was managed on a sprint-by-sprint basis using a 5x5 likelihood/impact scoring model, producing ten tracked risks across the project's four delivery sprints, with a combined project risk weight of 109.

15.1 Risk Scoring & Governance Guide

Scale	Likelihood	Impact Severity	Score Range	Severity Level
1	Very Low (10%)	Negligible	1−04	Low
2	Low (30%)	Minor	5−8	Low--Medium
3	Medium (50%)	Moderate	9−12	Medium--High
4	High (70%)	Major	15−19	High / Critical
5	Very High (90%)	Critical	20−25	Critical

Table 15.1 — Risk scoring scale and required governance action by severity band. Scores 13–14 are treated as Medium–High.

15.2 Sprint-Wise Risk Exposure

Sprint	Module / Phase	Risks (ID (Score))	Total Weight	Flag
Sprint 1	Planning & Design	RSK-01 (12), RSK-02 (6)	18	
Sprint 2	Customer & Merchant Hub	RSK-03 (20), RSK-04 (8), RSK-05 (16)	44	CRITICAL SPRINT
Sprint 3	NGO & Admin Hub	RSK-06 (9), RSK-07 (12), RSK-08 (8)	29	
Sprint 4	Testing & Deployment	RSK-09 (12), RSK-10 (6)	18	

Table 15.2 — Sprint 2 (Customer & Merchant Hub) is flagged as the project's critical sprint, carrying the highest cumulative risk weight (44).

15.3 Risk Register

ID	Sprint	Risk Description	L	I	Score	Severity
RSK-01	S1	Requirements gathered from sample restaurants/consumers fail to validate the loss-aversion and trust assumptions the design relies on	4	3	12	Medium-High
RSK-02	S1	The flat-fee, subscription, and price-floor structure locked in at Kickoff proves financially unviable once real transaction data is available	2	3	6	Low-Medium
RSK-03	S2	A food-safety incident traced to a surplus listing damages platform trust and brand, potentially irreversibly	4	5	20	Critical
RSK-04	S2	Payment gateway integration fails or has a high	2	4	8	Low--Medium

		transaction-failure rate at checkout				
RSK-05	S2	AI waste-prediction engine and dynamic pricing ladder produce inaccurate forecasts or mispriced listings	4	4	16	High / Critical
RSK-06	S3	Partner NGOs fail to collect donated food within the pickup window, causing spoilage	3	3	9	Medium--High
RSK-07	S3	Flat-fee billing and subscription engine miscalculates charges, causing revenue leakage or restaurant disputes	3	4	12	Medium--High
RSK-08	S3	User or restaurant data breach due to inadequate security controls in the Admin Hub	2	4	8	Low--Medium
RSK-09	S4	Integrating all four hubs surfaces critical bugs late in the cycle, close to launch	4	3	12	Medium--High
RSK-10	S4	Restaurants resist the phased, donation-first onboarding sequence at rollout	2	3	6	Low--Medium

Table 15.3 — Full risk register (L = Likelihood, I = Impact, both on a 1–5 scale; Score = L × I).

15.4 Risk Response Plans

ID	Response Plan	Owner	Status
RSK-01	Expand the interview sample before design sign-off; re-validate assumptions with a second round if results are inconclusive	Project Lead	Open
RSK-02	Treat the revenue model as a hypothesis; revisit commission/fee	Project Lead	Open

	levels after the Sprint 2 pilot with real restaurant data		
RSK-03	Strict hygiene/freshness badge verification, visible 'made at' timestamps, liability insurance, and a rapid incident-response protocol	Ops Lead	Open
RSK-04	Integrate at least two gateway providers, run full sandbox/UAT testing, add cash-on-pickup as a fallback	Backend Lead	Open
RSK-05	Rule-based fallback for new restaurants until 30+ days of data accumulate; enforce merchant price floor as a hard constraint	AI/Backend Lead	Open
RSK-06	Formal SLA with NGOs, backup NGO partners on standby, real-time pickup notifications	Ops Lead	Open
RSK-07	Reconciliation testing against manual calculations before go-live; audit log for every billing event	Backend Lead	Open
RSK-08	End-to-end encryption, role-based access control, and a security review before this module is signed off	Backend Lead	Open
RSK-09	Run incremental integration tests after each module's sign-off rather than one big-bang integration	QA Lead	Open
RSK-10	Communicate the long-term revenue-recovery benefit clearly; offer incentives to early adopters	Business Dev Lead	Open

Table 15.4 — Planned response/mitigation for each registered risk.

16. Cost Analysis & Budget

Labor cost was estimated by applying role-based daily rates (BDT/day) to the duration of each WBS work package, sprint by sprint. A risk-linked contingency reserve was then layered on top of labor cost, sized to each sprint's relative risk exposure.

16.1 Sprint 1 — Planning & Design (Total: 606,500 BDT)

WBS ID	Task	Duration	Role	Rate/day	Cost (BDT)
1.1	Kickoff meeting	1d	PM	4500	4500
1.2	Requirements gathering & interviews	14d	BA	3200	44800
1.3	Revenue & pricing model	12d	PM	4500	54000
1.4	Project plan sign-off	30d	PM	4500	135000
2.1	Wireframes & UI/UX design	28d	UX	2500	70000
2.2	Database schema design	5d	Dev Lead	3200	16000
2.3	System architecture & API design	21d	Dev Lead	3200	67200
2.4	Trust & freshness badge system spec	12d	Dev Lead	3200	38400
2.5	Dynamic pricing engine design	13d	Dev Lead	3200	41600
2.6	Design review & sign-off	30d	PM	4500	135000
	Sprint Total				**606,500 BDT**

16.2 Sprint 2 — Customer & Restaurant Hub (Total: 920,000 BDT)

WBS ID	Task	Duration	Role	Rate/day	Cost (BDT)
3.1	Registration & login	15d	Dev A	2500	37500

3.2	Search & discovery	16d	Dev A	2500	40000
3.3	Reservation & Surprise Food Box	15d	Dev A	2500	37500
3.4	Online payment integration	22d	Dev A	2500	55000
3.5	QR pickup verification	12d	Dev A	2500	30000
3.6	Ratings & reviews	9d	Dev A	2500	22500
3.7	Carbon impact dashboard	9d	Dev A	2500	22500
3.8	Student Saver Membership	9d	Dev A	2500	22500
3.9	Unit testing: Customer Hub	15d	QA	2500	37500
3.10	Module A review & sign-off	30d	PM	4500	135000
4.1	Restaurant registration & verification	17d	Dev B	2500	42500
4.2	Surplus listing upload	15d	Dev B	2500	37500
4.3	AI waste prediction engine	29d	Dev B	2500	72500
4.4	Dynamic pricing engine	20d	Dev B	2500	50000
4.5	Restaurant analytics dashboard	19d	Dev B	2500	47500
4.6	Sales & earnings reports	16d	Dev B	2500	40000
4.7	Unit testing: Restaurant Hub	15d	QA	2500	37500
4.8	Module B review & sign-off	30d	PM	4500	135000

	Sprint Total				**920,000 BDT**

16.3 Sprint 3 — NGO & Admin Hub (Total: 742,500 BDT)

WBS ID	Task	Duration	Role	Rate/day	Cost (BDT)
5.1	NGO partner onboarding & verification	12d	Dev C	2500	30000
5.2	Donation request workflow	15d	Dev C	2500	37500
5.3	Pickup scheduling	10d	Dev C	2500	25000
5.4	Food distribution tracking	10d	Dev C	2500	25000
5.5	Unit testing: NGO workflow	12d	QA	2500	30000
5.6	Module C review & sign-off	30d	PM	4500	135000
6.1	User & restaurant management console	17d	Dev D	2500	42500
6.2	Platform-wide order management	15d	Dev D	2500	37500
6.3	Flat-fee billing & subscription engine	21d	Dev D	2500	52500
6.4	Featured listing placement engine	14d	Dev D	2500	35000
6.5	Sustainability reports	14d	Dev D	2500	35000
6.6	System monitoring dashboard	12d	Dev D	2500	30000
6.7	Platform-wide analytics dashboard	18d	Dev D	2500	45000

6.8	Unit testing: Admin Hub	15d	QA	2500	37500
6.9	Module D review & sign-off	30d	PM	4500	135000
	Sprint Total				**742,500 BDT**

16.4 Sprint 4 — Integration & Deployment (Total: 639,000 BDT)

WBS ID	Task	Duration	Role	Rate/day	Cost (BDT)
7.1	Integrate all modules	24d	All Devs (A–D)	2,500 × 4	240000
7.2	System testing	25d	QA	2500	62500
7.3	Security & payment testing	20d	QA	2500	50000
7.4	User acceptance testing (UAT)	30d	Client	0	0
8.1	Deploy to production	12d	DevOps	2500	30000
8.2	Restaurant onboarding & training	15d	PM	4500	67500
8.3	Documentation & handover	12d	PM	4500	54000
8.4	Project closure	30d	PM	4500	135000
	Sprint Total				**639,000 BDT**

16.5 Risk-Linked Contingency Reserve

A contingency reserve was calculated per sprint as a percentage of labor cost, proportional to that sprint's share of total project risk weight (risk ratio = sprint risk weight ÷ maximum possible weight for that sprint's risk count).

Sprint	Labor Cost (BDT)	Risk Ratio	Contingency %	Reserve (BDT)

Sprint 1	606500	22/50 = 0.44	11%	66715
Sprint 2	920000	44/75 = 0.59	15%	138000
Sprint 3	742500	33/75 = 0.44	11%	81675
Sprint 4	639000	26/50 = 0.52	13%	83070
TOTAL	2908000	—	—	369460

Table 16.5 — Contingency reserve by sprint. Total reserve: 369,460 BDT.

16.6 Total Project Budget

Cost Component	**Amount (BDT)**
Total Labor Cost (all 4 sprints)	2,908,000
Risk-Linked Contingency Reserve	369,460
Estimated Total Project Cost	3,277,460

Table 16.6 — Estimated total project cost (labor + contingency). Client-side User Acceptance Testing (Task 7.4) is cost at 0 BDT as it is performed by the client/stakeholder, not billed labor.

17. Revenue Model

ReBite's revenue model was deliberately diversified across five streams, reducing dependency on any single monetization mechanism while keeping restaurant participation costs low and predictable.

- **Commission per Order:** A percentage-based commission (5%–10%) on each completed order, plus a flat platform fee (15 BDT/order) for cost predictability.
- **Restaurant Premium Subscription:** Paid tier offering restaurants deeper analytics, sales/earnings reports, and priority support.
- **Student Membership:** Recurring subscription revenue from students in exchange for additional discounts, priority reservations, and campus-exclusive offers.
- **Featured Listings:** Paid promotional placement for restaurant listings within the marketplace.
- **Sustainability Reports:** A B2B paid product offering restaurants and partners formal sustainability/CSR impact reports.

Per the project's own risk register (RSK-02), the flat-fee, subscription, and price-floor structure is treated as a hypothesis to be revisited once real transaction data is available from the Sprint 2 pilot, rather than a fixed, final pricing decision.

18. Feedback & Design Iterations

The project plan was revised iteratively in response to instructor and stakeholder feedback. The key feedback items and the resulting changes are summarized below.

Proposal — Consumer Psychology

Why should a consumer buy leftover food?

- **Change Made:** Reframed the offering around the term 'meals rescued' and emphasized sourcing from verified restaurants to build trust in food safety.

Proposal — Restaurant Psychology

Why should a restaurant give a discount when it can sell at the original price?

- **Change Made:** Repositioned the value proposition: without ReBite, unsold food yields 0 BDT and is thrown away; selling at a discount recovers real revenue instead of none.

Proposal — Revenue

How will the application earn revenue?

- **Change Made:** Added detailed analytics reports as a paid subscription feature, plus a flat platform fee of 15 BDT per order.

WBS — Depth

There should be more planning details.

- **Change Made:** Added a Level 4 breakdown beneath every Level 3 work package, enabling realistic estimation, assignment, and tracking.

WBS — Durations

Timings were not realistic.

- **Change Made:** Sign-offs were extended from 1 day to a 30-day review/approval window; total project duration is now approximately 16 months of active work (889 calendar days end-to-end).

WBS — NGO Structure

How should the NGO function be represented?

- **Change Made:** Changed from a standalone NGO Hub to an admin-managed NGO/Donation Workflow, matching the actual prototype structure.

19. Conclusion

ReBite is more than a food marketplace — it is a technology-driven sustainability platform that combines affordability, artificial intelligence, environmental responsibility, and social impact to create value for businesses, consumers, and society. This report has consolidated the project's proposal, requirements, stakeholder landscape, SWOT position, a four-level Work Breakdown Structure with a realistic ~16-month execution timeline, a sprint-based risk register covering ten identified risks, and a labor-cost and contingency budget of approximately BDT 3,277,460.

The plan reflects multiple rounds of feedback-driven revision — from consumer and restaurant psychology framing, to a diversified revenue model, to a WBS deep enough to support real estimation and tracking. With Sprint 2 (Customer & Restaurant Hub) identified as the project's highest-risk phase, and mitigation plans defined for every registered risk, ReBite is positioned as a governance-ready, execution-ready project for further development toward a working prototype and eventual market launch.